

Study guide

This guide describes the course organization and gives information on the Intended Learning Outcomes and the grading system applied.

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1) Summary

TGWK12 - Web Development Fundamentals	7.5 credits – First cycle university education
Coordinator: Jérôme Landré	Examiner: Jérôme Landré
Digital education platform : Canvas	Jönköping University – Autumn 2025

2) Content of the course

This course is dedicated to the **fundamentals building blocks of Web Development**. The web is based on the HTTP protocol that defines the way a request (for a resource: file or data) is sent from the client to the server and how the server will send back a response to the client with the resource or with an error message if there was a problem (see figure 1).

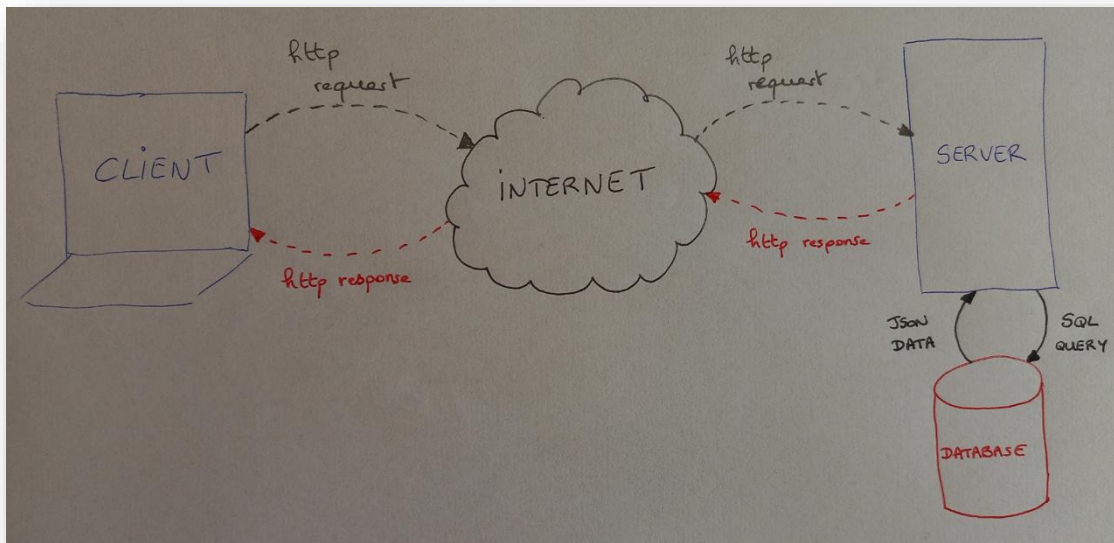


Figure 1: HTTP and the Web.

The Web architecture involves/uses many technologies/tools:

Internet (TCP/IP, IPv4, IPv6), a web server (HTTP server), a web client (browser), HTTP, a database, SQL, JSON, a programming language on the server (PHP, Node.js, ASP.net, Java, Ruby, Python, ...), description languages on the client (HTML, CSS) and a programming language on the client (Javascript).

In this course, we will look at all these different building blocks to produce web applications.

You will learn:

- Client-server communication and HTTP
- HTML
- CSS and CSS frameworks
- JavaScript (both Client-side and Server-side)
- Node.js and Node Package Manager
- Express
- Handlebars
- Git and Github
- Frameworks and related techniques
- Forms, cookies, and sessions
- Security

“More specifically you will learn how to create a web application to which a single user (or two: you and the teacher) can log in to modify the resources on it, such as blog posts or guestbook posts. To accomplish this your web application will use an SQLite3 database to store the resources and the web framework Express (with Node.js) to handle incoming HTTP requests. To avoid spending too much time on the graphical design of the web application you will also learn how to use a CSS framework with pre-styled graphical user interface components you can use. Your web application must, of course, also be secure, so you will also learn about common vulnerabilities in web applications and how you can prevent them.” Original text from Peter LARSSON-GREEN.

3) Intended Learning Outcomes

The intended learning outcomes of the course are found in the Course Syllabus.

ILOs	Evaluation
Knowledge and understanding	
- display knowledge of the fundamental building blocks of the web (HTTP and HTML) - demonstrate comprehension of common security concerns in web applications	Written Exam + Project Work
Skills and abilities	
- demonstrate the ability to construct web pages using HTML5, CSS, and CSS frameworks - demonstrate skills of programming in JavaScript - demonstrate skills of creating web pages using Node.js - demonstrate skills of constructing database-backed web applications	Written Exam + Project Work

4) Learning Activities

The teachers want the course, especially the labs, to be **progressive** so that you will learn all the building blocks one after another and apply the theoretical knowledge to a practical project.

a) Lectures

You will have many academic lectures that will present theoretical and practical knowledge on campus at the university.

b) Workshops/Seminars

To help you learn programming, there will be a lecture and two workshops on Databases for NMD students only. There will be two lectures on HTML/CSS, one lecture on Javascript and git, and one workshop on Javascript for DMP/NICS students only. And there will be a mixed workshop with NMD and DMP students on the overall project development.

To help you write code (Javascript, HTML, CSS), there will be live coding sessions which will focus on the demonstration about how to use Node.js to start, develop, and finish your project.

c) Lab Sessions

Each week you may attend a 4 hours lab session. At the lab session **you may ask** the teacher there for help and feedback with your project work.

There are 6 different lab sessions each week, but to avoid overfull rooms you may only join **one of them**, the one that has been given to you. A change is possible but you have to find another student that wants to swap groups with you. Both of you must send an e-mail to "jerome.landre@ju.se" to apply the change.

5) Examination Tests

The course contains two examination tests:

1. Written Examination (Inspira on campus online test): **2 credits**, graded U, 3, 4, and 5.
2. Project Work (creation of a dynamic web site): **5.5 credits**, graded U, 3, 4 and 5.

6) Grading guide

To receive a final grade on the course you need to pass both examination tests (Written Examination and Project Work). The project work is **individual OR by groups of TWO students**.

1. Written Examination: it is an Inspira online on campus examination that is mandatory. The questions will be graded on 100 points. The final grade will be given according to the scale:

<50	>=50	>=75	>=90
Fail	3	4	5

2. Project Work: it consists in creating a dynamic website with Node.js with Express and Handlebars, serving dynamic web pages (HTML, CSS) where data come from a database (SQLite3).

The code will be graded by Jérôme LANDRÉ, 3, 4, and 5, or fail (not a good idea to get this one) according to the document:

Grading criteria for the Web Development Fundamentals project – 2025 – v1.2

Grade	Requirement/Code content for INDIVIDUAL project	Requirement/Code content for GROUPS of 2 students project
3	- The code works without error, there are sufficient comments in the code, - The code uses Node.js, Express, Handlebars and SQLite3, - The final website is a one-page dynamic application, the menu MUST contain at least: Home, List (can be renamed depending on your project choice), About, Contact, - The Database has at least two tables (at least 5 records) that are displayed on the pages, the tables have at least 3 fields, - The menu "List" must link to a page displaying the list of elements (except the user table used for grade 5) from one of your tables, a detailed view of one element is given when you click on it, - Basic security (no SQL injection) and login/encrypted password checked to modify the data, there is a message to know if the user is connected or not, - The login and password for the user are respectively "admin" and "wdf#2025", - Usage of SELECT/INSERT/UPDATE/DELETE (CRUD) operations to display/modify one table, - Acceptable UI/UX using basic CSS.	Requirements are the same as the left column - WITH three major changes in red: - The Database has at least four tables (at least 10 records) that are displayed on the pages, the tables have at least 3 fields, - Usage of SELECT/INSERT/UPDATE/DELETE (CRUD) operations to display/modify two tables,
4	- Requirements for grade 3 above PLUS: - Database with three or more tables (at least 5 records in each table), two of them being linked (SQL INNER JOIN) in at least one query (that is used in the pages), - There is a list with a dynamic pagination system (at least 3 pages with 3 elements per page) for one table (so, at least 9 records in this table) - Moderate UI/UX with Flexbox or Grid CSS.	Requirements are the same as the left column and the cell above WITH three major changes in red: - Database with five or more tables (at least 10 records in each table), three of them being linked (SQL INNER JOIN) in at least one query (that is used in the pages),
5	- Requirement for grade 4 above PLUS: - Create a complete user management system: create new user with hashed password, listing, updating, and deleting users (SELECT/INSERT/UPDATE/DELETE in your users' table), the users can log in on your website, - Advanced UI/UX being fully responsive (computer, tablet, and phone) with Flexbox or Grid CSS or a CSS Framework.	Requirements are the same as the left column and the cell above WITH two major additions in red: - In one of the forms, there is a possibility to upload an image (PNG or JPG) in a directory on the server with SELECT/INSERT/UPDATE/DELETE on this image which is used and displayed in the web pages,

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3. The students need to pass Written examination **AND** Project work to be graded. The **final grade of the course** will be given according to a mix in between grades from 1. and 2.

7) Re-examination

On the picture below, the study and exam periods during the academic year 2025-2026 are presented.

Source: <https://ju.se/student/en/studies/academic-calendar/school-of-engineering.html>

Exam 1: week 43, 2025

Exam 2: week 3, 2026 (re-exam 1)

Exam 3: week 23-24, 2026 (re-exam 2)

The academic year 2025/2026

Autumn 2025 (2025-09-01 – 2026-01-18)

Re-exam (per 3-4) 250818-250824 v.34

Introduction week 250825-250831 v.35

Study period 1 250901-251017 v.36-42

Exam period 251018-251026 v.43

Study period 2 251027-251212 v.44-50

Exam period 251213-251221 v.51

Re-exam period (per 1+4) 260110-260118 v.3

Spring 2026 (2026-01-19 – 2026-06-07)

Study period 3 260119-260313 v.4-11

Re-exam period (per 2) 260207-260215 v.7

Exam period 260314-260322 v.12

Study period 4 260323-260522 v.13-21

Exam period 260523-260531 v.22

Graduation 260529

Re-exam period (per 1-3) 260601-260610 v.23-24

Re-examination for the Written examination and the Project Work are offered at the re-exam periods in January and June. If you need to set a precise date and time in advance, email the course coordinator at jerome.landre@ju.se as soon as possible. **The project work will only be graded U/3 at re-examination.**

If you haven't passed some examination tests after the three examination opportunities (ordinary + 2 re-exams) you need to re-take those tests the next time the course runs. Note that the instructions for those tests might have changed the next time the course runs and that you should take them according to the new instructions.

8) Course Literature and Other Teaching Aids

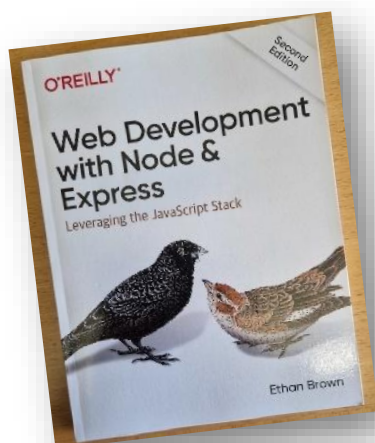
The course is not based on a specific book but on many freely available resources on the Internet. This section gives several of them:

- **HTTP:**
 - o Introduction to HTTP - <https://launchschool.com/books/http> by Launch School
 - o TutorialsPoint: <https://www.tutorialspoint.com/http>
 - o Mozilla Developer Network (MDN): <https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview>
- **HTML & CSS:**
 - o MarkSheet <https://marksheet.io/> by Jeremy Thomas
 - o Mozilla Developer Network (MDN): <https://developer.mozilla.org/en-US/docs/Web/HTML>, <https://developer.mozilla.org/en-US/docs/Web/CSS>
 - o W3Schools: <https://www.w3schools.com/html/>
 - o CSS tricks: <https://css-tricks.com/>
 - o The Spectre framework: <https://picturepan2.github.io/spectre/index.html>
- **Javascript:**
 - o Client side: Eloquent JavaScript 3rd edition <https://eloquentjavascript.net/> by Marijn Haverbeke
 - o Server-side: Node.js documentation, <https://nodejs.org>
- **SQLite3:**
 - o Getting started with SQLite <https://www.bu.edu/csmet/files/2021/03/Getting-Started-with-SQLite.pdf> by Warren Mansur
 - o SQLite with Node.js tutorial <https://www.sqlitetutorial.net/sqlite-nodejs/> by sqlite tutorial

If you need to read more, I suggest (**not mandatory**) that you look at the book (available at JU library):

“Web Development with Node & Express”, second edition, by Ethan BROWN,
publisher: O'Reilly, 2020, ISBN: 978-1-492-05351-4

Be careful, this book covers more (Quality Assurance, MongoDB, ...) than the course and not fully (SQLite3 is not used in this book). So, interesting chapters are: 1, 2, 3, 4, 6, 7, 8, 9, 10 and page 179 for route parameters.



9) Time Plan

Course organisation

Week	36	37	38	39	40	41	42	43
Programme	1 DMP/NICS	2 DMP/NICS	3	4	5	6	7	8
Lecture 1 1:45	Organisation & HTTP Jérôme	Javascript (client-side) & Git Jérôme	Databases 2 Jérôme	Node.js + Express Jérôme	Express + REST API Jérôme	Forms/authentication/ session/cookies Jérôme	Security/AI tools Jérôme	Advanced Code 1 (pagination) Jérôme
Lecture 2 1:45	HTML & CSS 1 Jérôme	Databases 1 Jérôme	Project Work Jérôme	Express/Handlebars Jérôme	Git/Databases Jérôme	Databases/CRUD Jérôme	Advanced Code 2 (upload images) Jérôme	Inspira online, on campus Examination Jérôme
Lecture 3 1:45	HTML & CSS 2 Jérôme							
Workshop (Active Learning Classroom) 1:45		Javascript (client-side) & Git Jérôme	Workshop Project Setup Jérôme					
Lab 3:45	Lab1 HTML/CSS ? + Mira + Jérôme	Lab2 Advanced CSS/JS ? + Mira + Jérôme	Lab3 Node/Express, Git ? + Mira + Jérôme	Lab4 Express/Handlebars & Project Work ? + Mira + Jérôme	Lab5 Databases & Project Work ? + Mira + Jérôme	Lab6 Forms, Sessions, Cookies & Project Work ? + Mira + Jérôme	Lab7 Pagination, Security, Full CRUD & Project Work ? + Mira + Jérôme	
Project	Portfolio or Open Project / Individual or groups of TWO students							
	Project evaluation Jérôme							

Number of students registered in the course: 160+

Number of lab groups: 6 in total, 2 for Mira, 1 for ?, 1 for ?

Number of needed lab assistant students: 2

People involved in the course:

?
?
Mira Pop
Jérôme Landré

1 lab per day, Thursdays and/or Fridays

1 lab per day, Thursdays and/or Fridays

1 lab per day, Thursdays and/or Fridays

The course has two 1:45 sessions a week (except first three weeks).

The course has one 3:45 lab session per week.

Evaluation: Inspira, project work evaluation